Data Pipeline Architecture

Stripe is a company that must adapt to handle a large volume of data on a global scale. The main challenge is connecting data of different types. This justifies the combination of various OLTP, OLAP, and NoSQL databases. After outlining the general architecture of these databases, this document describes the different techniques for processing, transforming, and synchronizing data.

1. Architecture Overview

This pipeline connects three types of databases, each with its own role. The first is an OLTP (Online Transaction Processing) database. It is optimized for everyday operations. Data flows in real time, and the queries that can be performed on it are simple and analyze small volumes of data. This system has ACID properties, which guarantee its reliability. ACID corresponds to four rules: A transaction is complete and cannot be altered. The database is said to be consistent, meaning it cannot violate its own rules. Two simultaneous transactions cannot interfere with each other, and a confirmed transaction remains valid even if the system crashes. This system uses PostgreSQL to function. These form the operational foundation of Stripe.

The second database is an OLAP (Online Analytical Processing) database. It is designed to analyze large volumes of data with resource-intensive queries. It uses Snowflake technology, a data warehouse optimized for analytical queries. Its organization is structured according to business questions to facilitate analysis. The chosen structure for Stripe is a star schema, which allows for fast data reading and aggregation.

The last database is a NoSQL database. This system uses MongoDB technology, which allows for the storage of both semi-structured and unstructured data that cannot be stored in other databases. It concerns, for example, user behavior, inputs for the machine learning model, and feedback from various customers. This data has no fixed structure; without NoSQL, it could not correspond to any other type of database.
To connect these databases, two types of processing coexist: real-time streaming and batch processing, which is the processing of data at regular intervals. Real-time processing is useful for sensitive data, such as fraud detection, error logs, or user interactions. Batch processing is used to populate the OLAP database nightly for analysis.

2. Real-time streaming via Apache Kafka

Apache Kafka is a distributed messaging system that allows for the real-time transport of events between different databases. It retrieves data from the OLTP system and immediately forwards the information to the OLAP systems on Snowflake RAW and NoSQL on MongoDB. More specifically, it sends error log data and user behavior data to MongoDB, and transaction data to Snowflake RAW. Apache Kafka technology ensures that no information is lost, even during traffic spikes.

3. Batch processing with Airbyte and Apache Airflow

Airbyte is an open-source tool that automates the extraction and loading of data between its source and destination. It extracts information from the OLTP server and loads it into Snowflake RAW, which corresponds to the OLAP database. This technology eliminates the need to worry about the connection between the different platforms, which are natively managed by Airbyte. This software works in tandem with Apache Airflow, which allows for the scheduling of these data exchanges. In the case of Stripe, these data transfers should ideally occur once per night, when traffic is lower. Airflow goes beyond simply scheduling data extraction. It also allows for scheduling the dbt transformation of this data, which involves cleaning it and building the fact and dimension tables for OLAP. It also allows for scheduling the calculation of materialized OLAP views, which correspond to frequently asked business questions. This accelerates analysis. It also orchestrates the synchronization from NoSQL to OLAP. Airflow technology allows for receiving an alert directly if a task is not automatically performed.

4. Data Transformation with dbt

dbt is an SQL transformation tool that allows you to take raw data from Snowflake RAW and transform it into an analytical schema in Snowflake Analytics. It also allows, among other things, version control and the automatic generation of documentation. Specifically, it builds OLAP fact tables such as fact_transaction, fact_fraud, and fact_compliance, as well as dimensions like dim_customer and dim_merchant. It also recalculates materialized views such as summary_daily_revenue and summary_fraud_by_country. This software also ensures data quality through automated testing.

5. Data Synchronization and Consistency with CDC

Change Data Capture is a technique that monitors and captures all changes occurring in a database in real time. It allows tracking every OLTP modification in real time. This enables updates to propagate to MongoDB and Snowflake, for example, if a customer changes their email address. This prevents inconsistencies between the three systems. Specifically, this is made possible by referencing customer_id and transaction_id in the NoSQL database. This creates a link between the different systems. In case of data conflicts, the OLTP is the authoritative source of truth.

This pipeline allows Stripe to process its data both operationally and analytically, combining structured and unstructured data. Kafka ensures the real-time transmission of critical data, such as fraud detection. Airbyte and Airflow handle the loading and orchestration of analytical data. Dbt transforms raw data into analytical schemas, and CDC ensures consistency across the three systems. Together, these tools form a robust, scalable architecture capable of meeting Stripe's growing needs.